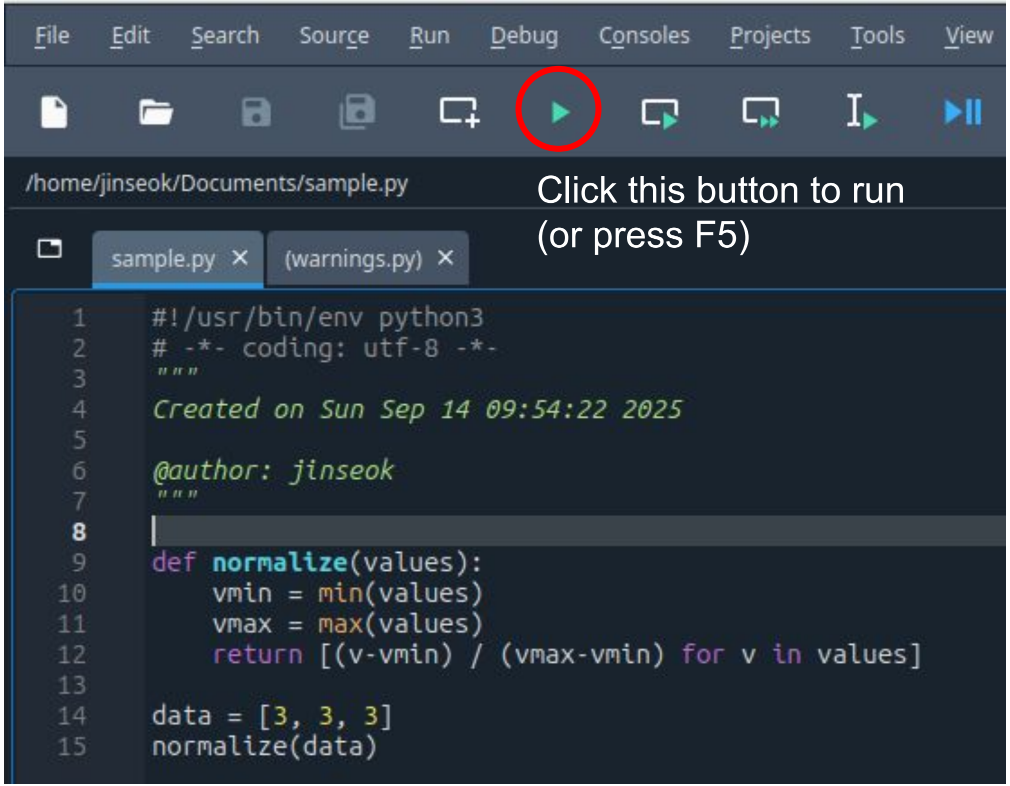
Spyder - simple debugging

ReproRehab Pod2

Running an erroneous file

Run buggy_script.py.



Running an erroneous file

You get the ZeroDivisionError: division by zero.

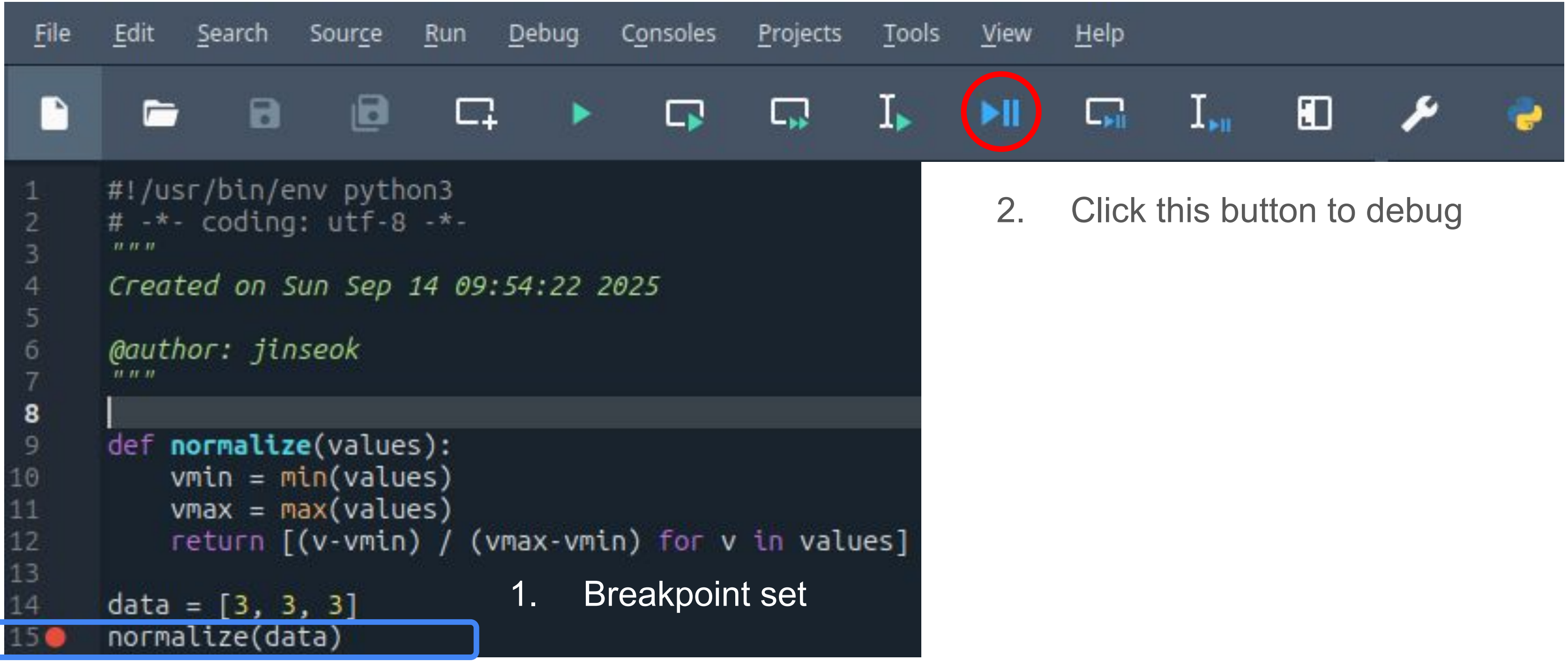
```
In [1]: %runfile /home/jinseok/Documents/sample.py --wdir
---------------------------------------------------------------------------
ZeroDivisionError                         Traceback (most recent call last)
File ~/Documents/sample.py:15
     12     return [(v-vmin) / (vmax-vmin) for v in values]
     14 data = [3, 3, 3]
---> 15 normalize(data)

File ~/Documents/sample.py:12, in normalize(values)
     10 vmin = min(values)
     11 vmax = max(values)
---> 12 return [(v-vmin) / (vmax-vmin) for v in values]

ZeroDivisionError: division by zero
```

Debugging - Step 1

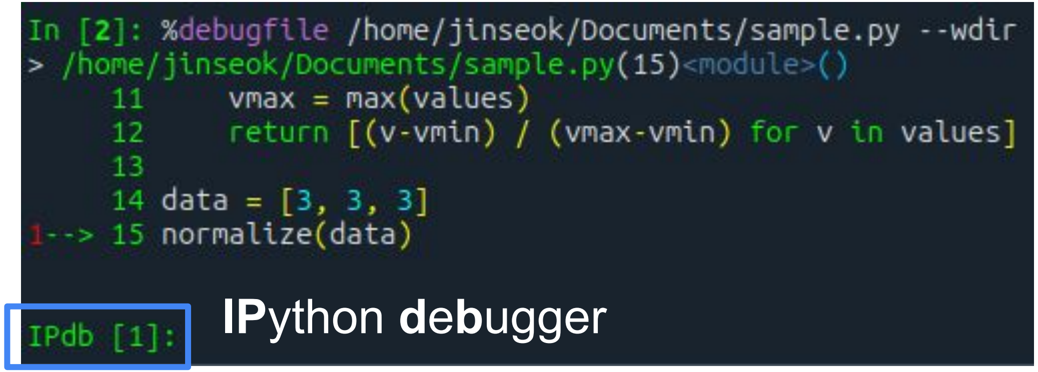
Set the breakpoint at the end of your script and debug the file (Ctrl+F5)



1. Breakpoint set
2. Click this button to debug

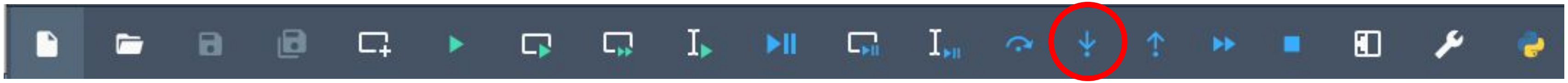
Debugging - Step 2

Debugging starts. It will run the script until line 14.



Blue arrow positioned at line 15

More menu icons are available. Step into a function (Ctrl + F11).



Click this icon once

Debugging - Step 3

You see the blue arrow now jumps to the first line of the function: normalize.

Then check ‘variable explorer’. You will now see a variable: **values**. This is the parameter of the function normalize.

At line 15, the variable: **data** is provided as **values** to the function[That’s why values is a list of size 3 ([3, 3, 3])

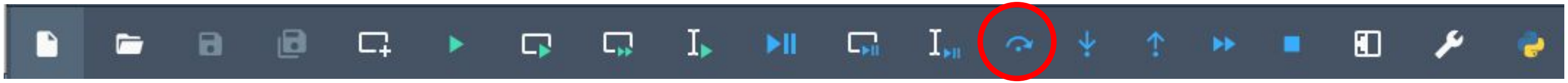
```
1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Created on Sun Sep 14 09:54:22 2025
5  
6  @author: jinseok
7  """
8  
9  def normalize(values):
10     vmin = min(values)
11     vmax = max(values)
12     return [(v-vmin) / (vmax-vmin) for v in values]
13 
14 data = [3, 3, 3]
15 normalize(data)
```

Name	Type	Size	Value
data	list	3	[3, 3, 3]
values	list	3	[3, 3, 3]

Debugging - Step 4(A)

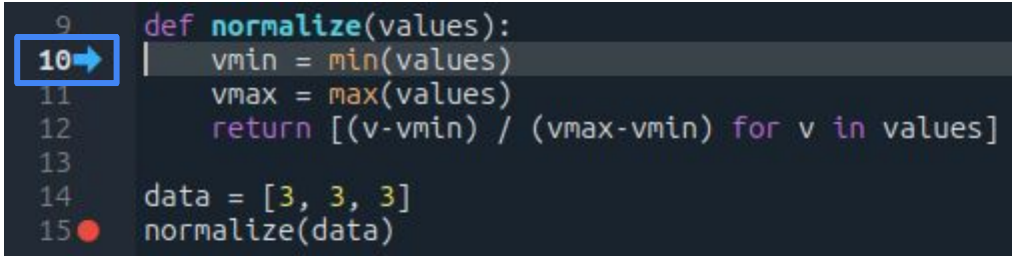
Click the curved arrow button (Debug current line, Ctrl + F10) once and check if the blue arrow moves down by one line.

Click it!



The blue arrow moves down by one line

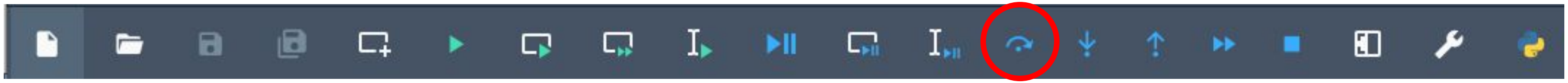
If the current line can run without any error, it will run and proceed.



Debugging - Step 5

Click the curved arrow button one more time and check the variable explorer.

Click it one more time



The blue arrow should be at line 11

```
9   def normalize(values):
10      vmin = min(values)
11→     vmax = max(values)
12      return [(v-vmin) / (vmax-vmin) for v in values]
13
```

min([3, 3, 3]) = 3

This is what's displayed in the console

```
IPdb [1]: !next
```

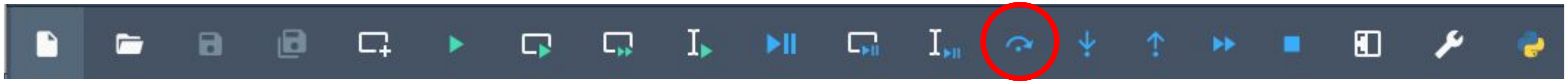
variable: **vmin** is generated

Name	Type	Size	Value
data	list	3	[3, 3, 3]
values	list	3	[3, 3, 3]
vmin	int	1	3

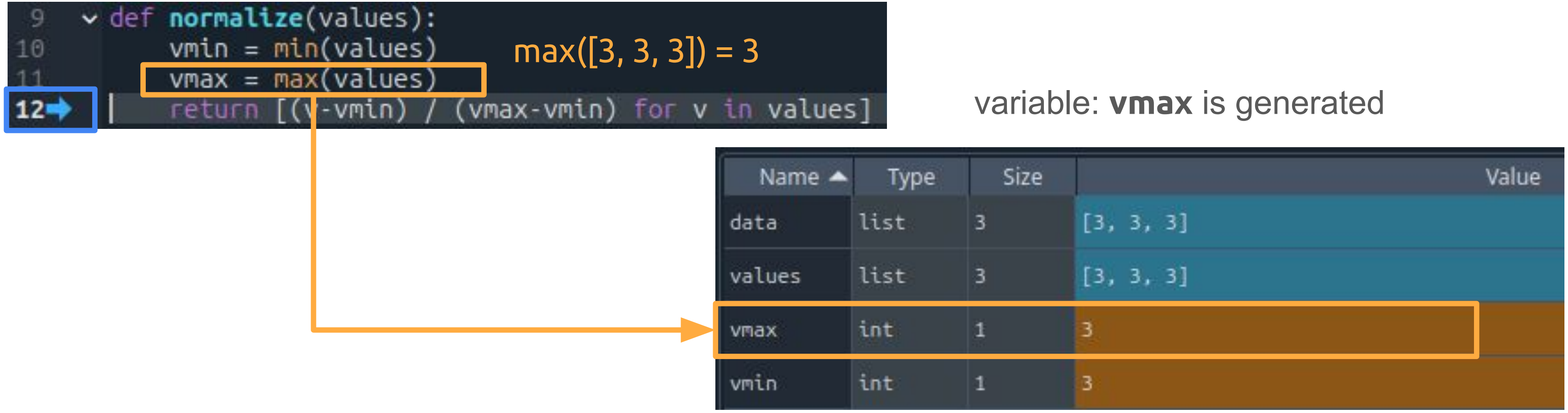
Debugging - Step 6

Click the curved arrow button one more time and check the variable explorer.

Click it one more time



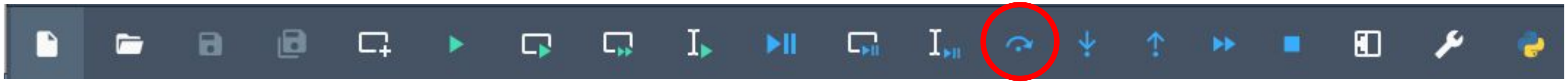
The blue arrow should be at line 12



Debugging - Step 7

Click the curved arrow button one more time and check the console window.

Click it one more time



Line 13 is returning ZeroDivisionError

```
IPdb [1]: !next
ZeroDivisionError: division by zero
```

```
12➡ |     return [(v-vmin) / (vmax-vmin) for v in values]
```

Name	Type	Size	Value
data	list	3	[3, 3, 3]
v	int	1	3
values	list	3	[3, 3, 3]
vmax	int	1	3
vmin	int	1	3

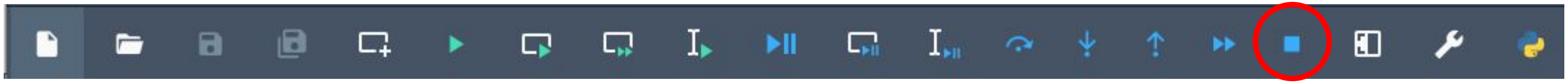
v is an element of **values** at each iteration. At the first iteration, **v** is values[0] = 3

(v-vmin) is (3-3) = 0.
(vmax-vmin) is (3-3) = 0.
Therefore, it is 0/0. Error!

Debugging - Step 8

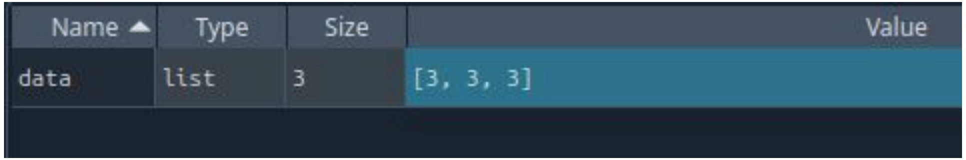
Once you pinpoint the cause of the error, you can stop debugging by pressing the stop button.

Click this to STOP



Your console window will no longer be in “IPdb”

Variables generated during debugging are also removed.

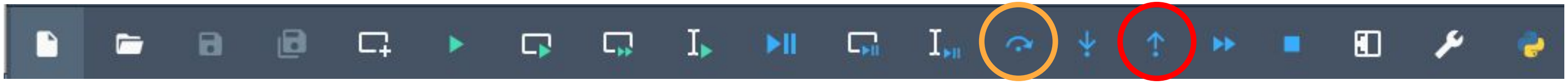


Name	Type	Size	Value
data	list	3	[3, 3, 3]

Debugging - Step 4(B)

Instead of clicking the curved arrow button multiple times as we did in steps 4A - 7, you can skip running individual lines that don’t raise any error. Click the upward pointing arrow (Execute until function returns, Ctrl + Shift + F11), and you will immediately be at step 7.

Click it!



This is the console window you will see. You can then click the **curved arrow button** once.